

Chronic kidney disease happens when the kidneys are damaged for more than a few months. The kidneys clean the blood by removing waste and extra fluid, which leaves the body as urine. They help control blood pressure and balance salt and minerals. The kidneys also help the body make red blood cells and keep bones strong.

In the early stages of chronic kidney disease, you might not feel sick or have any symptoms. You might not know that you have kidney disease until the condition is advanced.

Healthcare professionals can find kidney disease through blood and urine tests. These tests look for high levels of waste in the blood or unusual things in the urine, such as protein or tiny amounts of blood. They also might use tests such as ultrasound or CT scans to look at the kidneys.

Treatment for chronic kidney disease aims to slow down damage to the kidneys. This often is done by managing the cause of the damage. But, even after a cause has been found and treated, kidney damage could still get worse. Chronic kidney disease can progress to end-stage kidney failure. This is fatal unless a person gets a kidney transplant or starts dialysis, a treatment that does the job of the kidneys.

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Symptoms

Chronic kidney disease symptoms tend to develop slowly over time.

Loss of kidney function can cause a buildup of fluid or waste in the body. It also can cause acids, potassium and phosphate to build up in the blood.

Symptoms of kidney disease often are vague. Most people with kidney disease have no symptoms until the very last stages of kidney disease. This is known as advanced kidney disease.

When chronic kidney disease becomes advanced, loss of kidney function can cause:

- Nausea.
- Vomiting.
- Loss of appetite.
- Fatigue and weakness.
- Sleep problems.
- Decreased mental sharpness.

- High blood pressure that's hard to manage.
- Shortness of breath, if fluid builds up in the lungs.

Stages of kidney disease

There are five stages of kidney disease. To learn what stage you have, your healthcare team performs a blood test to check how well the kidneys are working. This blood test is called the estimated glomerular filtration rate (eGFR). The eGFR measures how much blood the kidneys filter each minute, recorded as milliliters per minute (mL/min). A low eGFR number means the kidneys are working less well.

When the kidneys no longer work at the level needed to keep a person alive, it's called end-stage kidney disease. This happens when you have an eGFR under 15.

As a part of kidney disease staging, your healthcare professional also might test whether you have protein in your urine.

Kidney disease stage	eGFR, mL/min	Kidney function
Stage 1	90 or above	Healthy kidney function
Stage 2	60 to 89	Mild loss of kidney function
Stage 3a	45 to 59	Mild to moderate loss of kidney function
Stage 3b	30 to 44	Moderate to severe loss of kidney function
Stage 4	15 to 29	Severe loss of kidney function
Stage 5	Less than 15	Kidney failure

Source: National Kidney Foundation

When to see a doctor

If you have an ongoing health condition that raises your risk of kidney disease, your main healthcare professional may want to monitor the health of your kidneys. Having long-standing high blood pressure, diabetes or an autoimmune disease such as lupus can cause damage to the kidneys.

Regular checkups to monitor how well the kidneys are working may include blood tests and tests to check for protein in the urine. You may see your main healthcare professional every

6 to 12 months for these tests. If test results show possible kidney problems, you may be referred to a doctor who specializes in kidney diseases. This type of doctor is called a nephrologist.